

Sampling distribution

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Definition of sampling distribution

- Theoretical distribution
- A series of statistic from independent random samples of *same size* from *same population*
- Shape and spread depends on sample size (n) and variation in parent population

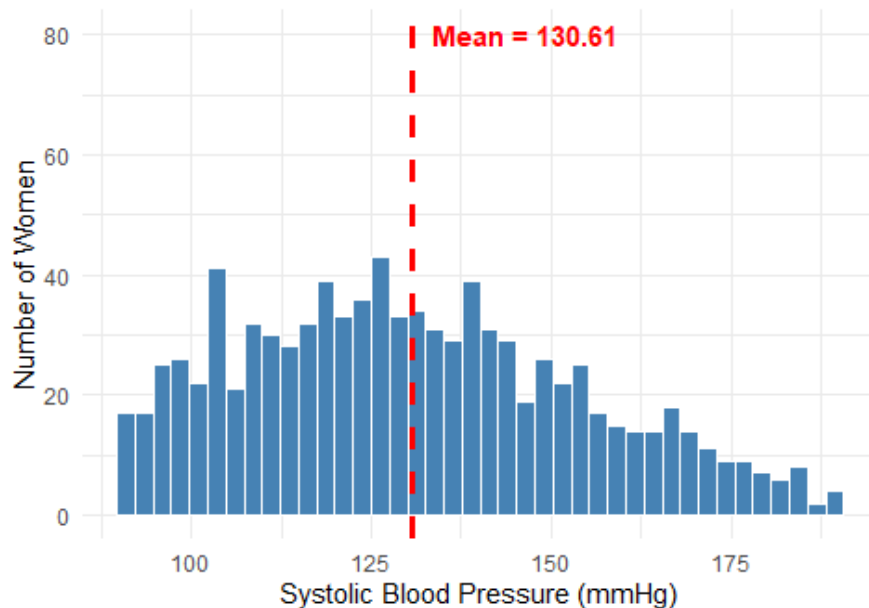
Hypothetical population data

- 1000 population of >50 years old women
- Gamma distribution to get a right skewed distribution
- Get the range of systolic blood pressure between 90mmHg and 190mmHg
- The population mean is 131.

Distribution of the data

Hypothetical Population: Systolic Blood Pressure

908 Women over 50 | Population Mean = 130.61 mmHg



Sampling

- From this population of 908 women, **1000 samples** will be taken 3 ways
 1. **5 women** in each sample
 2. **30 women** in each sample
 3. **100 women** in each sample
- Then, we calculate the **sample statistic(mean)** of each sample and distribute along the X axis

Sample Statistics

Group	Size	Mean	Median	SD	Q1	Q3	Min	Max
1. Population	908	130.6	128.2	23.5	112.0	147.1	90.0	188.5
2. Sample Means (n=5)	1000	130.5	129.9	10.4	123.3	137.5	98.7	168.3
3. Sample Means (n=30)	1000	130.6	130.4	4.2	127.9	133.3	116.6	145.2
4. Sample Means (n=100)	1000	130.6	130.7	2.4	129.0	132.2	123.9	137.7

Sampling Distribution

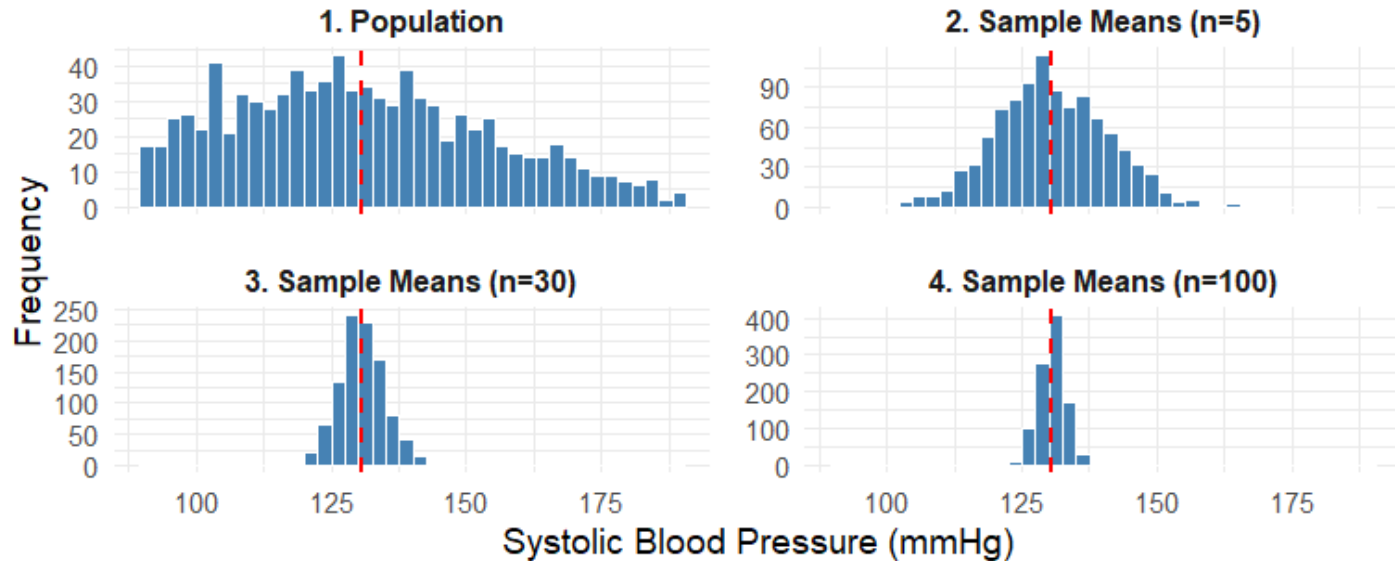
- When the sample size is large enough
- Distribution of the sample means follows **Normal distribution**
- **Mean of the samples' means** is equal to **True Population mean** ($\bar{X} = \mu$)

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Distribution of Sample means

Demonstrating the Central Limit Theorem

Red dashed line = True Population Mean (130.61 mmHg)



Distribution of the estimates

- *Mean of Sample means*($\bar{X}$)= 131
- Standard error of the mean (**SEM**) measures the variability of the distribution
- If population SD is known, $SEM = \frac{\sigma}{\sqrt{n}}$
- if only one sample SD is known, $SEM = \frac{s}{\sqrt{n}}$
- Large *SEM* - Imprecise estimate
- Small *SEM* - Precise estimate

Standard error of the estimate

ID	N	Mean	SEM
Population	908	130.6	0.8
1	5	131.3	12.4
2	30	136.8	3.6
3	100	130.2	2.4
4	500	129.9	1.0

Central limits theorem

- As sample size increases, the distribution of sample means approaches a normal curve regardless of the original population's shape.
- The sampling distribution stays centred at the true population mean and its variance is $\frac{\text{population variance}}{\text{sample size}}$
- Standard error decreases as sample size grows
- The Central Limit Theorem and sampling distribution are foundation for inferential statistics.

Use in inferential statistics

- **Confidence Interval (CI):**

$$\text{Estimator} \pm \text{Reliability Coefficient} \times \text{Standard Error}$$

- **Test Statistic (for P-values):**

$$\frac{\text{Relevant Statistic} - \text{Hypothesized Parameter}}{\text{Standard error}}$$

THANK YOU